

Computing & Numerical Methods

Example C++ Test

50 minutes

Instructions

Please read the entire Task before starting.

You should write a single C++ program to address the task below.

All code should be contained in a single `.cpp` file.

Your code **must** be C++11-standard compliant. You must **not** use any other header files beyond the standard C++ headers. These requirements will automatically be satisfied if you only use the language capabilities discussed in the course.

Your code (the `.cpp` file) should be submitted to Blackboard Learn by the end of the test.

Task

The functions $\sin(x)$ and $\cos(x)$ can be approximated by the following truncated series expansions:

$$\sin(x) \approx \sum_{k=0}^K \frac{(-1)^k x^{2k+1}}{(2k+1)!}, \quad \cos(x) \approx \sum_{k=0}^K \frac{(-1)^k x^{2k}}{(2k)!}. \quad (1)$$

Write a single program which requests the value of x from the user, with x limited to the range $-\pi < x < \pi$, and computes the **absolute error** in approximating

$$y = \sin^2(x) + \cos^2(x). \quad (2)$$

You should:

1. Write a function which calculates the factorial of a positive integer. [10%]
2. Write functions which compute the approximations to $\sin(x)$ and $\cos(x)$, given parameters x and K . [40%]
3. Request the floating-point value x (in radians) from the user. Check it is valid and in the correct range. [15%]
4. Using this value of x , calculate:
 - the true values of $\sin(x)$ and $\cos(x)$ using the functions provided by C++;
 - the approximation of $\sin(x)$ and $\cos(x)$ calculated using the Equations (1) and with $K = 5$;
 - the **absolute error** in the approximation of $y = \sin^2(x) + \cos^2(x)$. [15%]
5. Use good readable coding style (spacing, indentation, comments, etc.). [20%]

END OF TEST